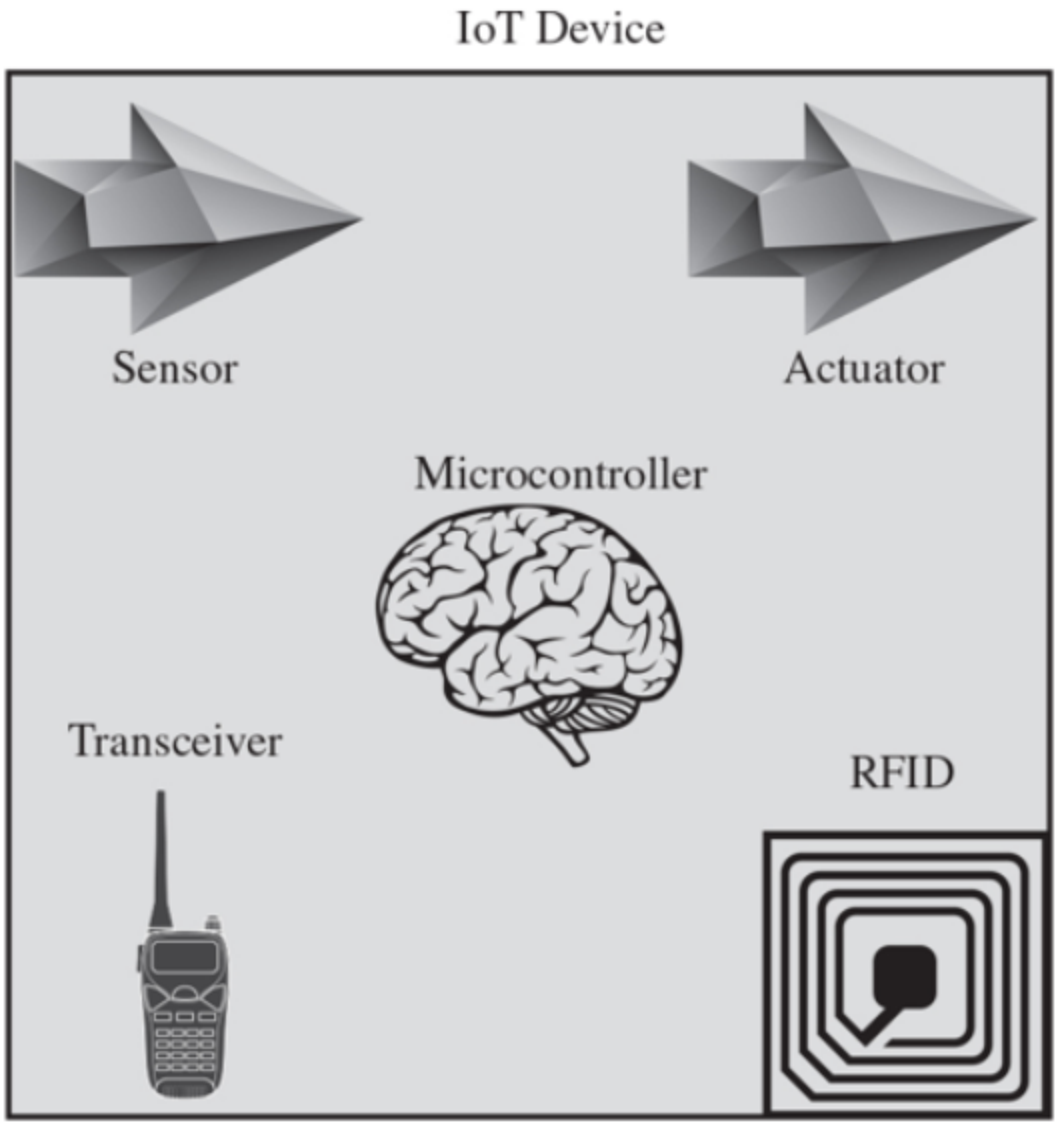
The Internet of Things

Components of IoT-enabled Devices:

- **Microcontroller.**
- **Sensors.**
- **Actuator.**
- **Transceiver.**
- **RFID-Technology.**
- **Communication/Messaging Protocols.**



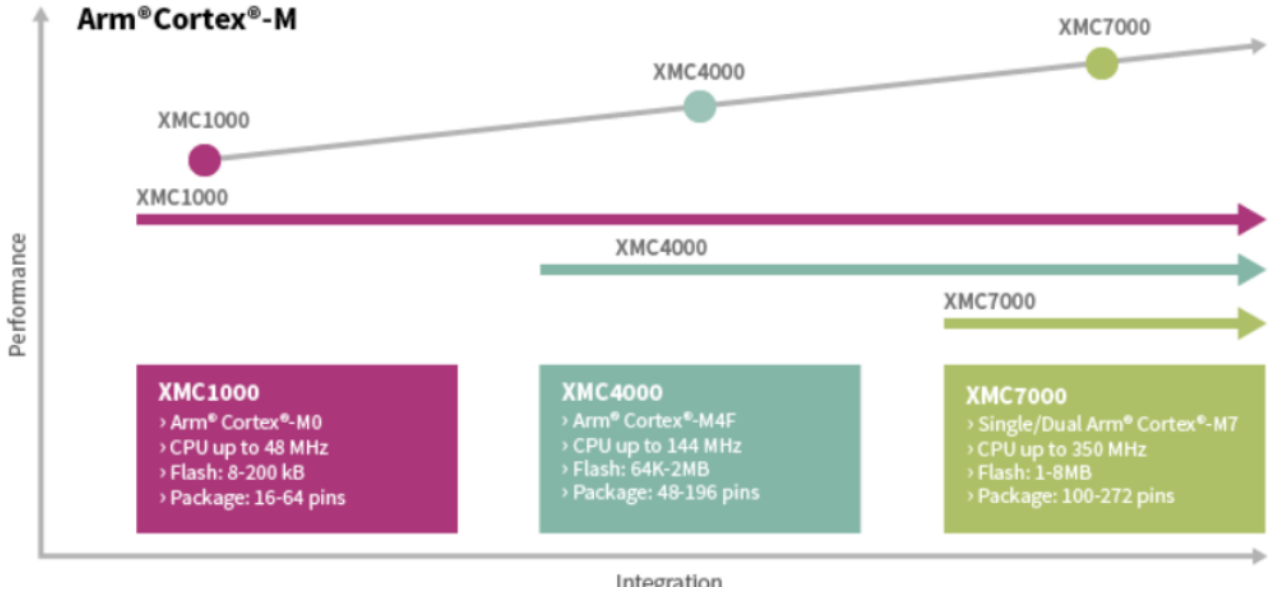
Transceiver (Transmitter/Receiver)

- WiFi.
- LTE.
- 5G.
- 5G+.

Transponder (Transmitter/Responder)

Microcontroller

- Raspberry Pi.
- Arm Cortex



RFID

Sensors

11:26 AM · Oct 30, 2021

In October 2020, we installed thermocouples and strain gauges in concrete slabs eastbound on Highway 407 ETR near the Pine Valley interchange. The sensors are currently collecting temperature and strain data within different depths for analysis to evaluate the effects of high temperatures on the performance of our concrete pavement. This will help us understand how significant fluctuations in temperatures contribute to the localized deterioration of our concrete surfaces.

https://www.407etr.com/documents/investors/2020SustainabilityReport.pdf (https://www.407etr.com/documents/investors/2020SustainabilityReport.pdf)

Actuator (Responder)

- Mechanical
- Electronic (Responder in Transponder)